Lab 2: CIFAR-10 Classification

Student ID: Student Name:

**Table 1. Performance on CIFAR-10**

| Model | Train Loss | Train Acc. | Test Loss | Test Acc. | Parameters |
|---|---|---|---|---|---|
| LinearClassifier | | | | | |
| MLP-512 | | | | | |
| MLP-2048 | | | | | |
| ConvNetV1 | | | | | |
| ConvNetV2 | | | | | |
| ConvNetV3-Reg | | | | | |
| ConvNetV3-Aug | | | | | |
| ResNet50-Scratch | | | | | |
| ResNet50-Finetune | | | | | |

**Problem 1: Linear Classifier**

A. Implement LinearClassifier class in src/models.py
B. Write the performance of LinearClassifier to Table 1
C. Write the number of trainable parameters of LinearClassifier to Table 1

**Problem 2: Multilayer Perceptron (MLP)**

A. Implement MLP class in src/models.py
   - MLP class has two hidden layers and one output layer
   - Each hidden layer consists of Linear and ReLU layers
   - Hidden layers should be grouped by nn.ModuleList
B. Run MLP-512 using configs/mlp_512.yaml
   - The dimension of hidden layers in MLP is set to 512 in the YAML config
C. Write the performance of MLP-512 to Table 1
D. Write the number of trainable parameters of MLP-512 to Table 1
E. Run MLP-2048 using configs/mlp_2048.yaml
   - The dimension of hidden layers in MLP is set to 2048 in the YAML config
F. Write the performance of MLP-2048 to Table 1
G. Write the number of trainable parameters of MLP-2048 to Table 1
H. Compare Linear, MLP-512, and MLP-2048. Which is the worst model? Which is the best model? Give a reason why the worst model provides lower test accuracy than the others? Give a reason why the best model provides higher test accuracy than the others?

**Problem 3: Convolutional Neural Network**

A. Implement ConvNetV1 class in src/models.py
   - ConvNetV1 class has four hidden layers and one output layer
   - Each hidden layer consists of Conv2d and ReLU layers
   - ConvNetV1 class has input arguments “dims” and “strides” are Python List to specify the input and output channels, and the strides of Conv2d layers.
   - All Conv2d layers work with 3x3 convolution kernels and padding 1.
   - Link the output of last hidden layer to the output layer using global average pooling
   - Hidden layers should be grouped by nn.ModuleList
B. See the model arguments in configs/convnet_v1.yaml. Predict the shape of the output tensor of each hidden layer
C. Write the performance of ConvNetV1 to Table 1
D. Write the number of trainable parameters of ConvNetV1 to Table 1
E. Compare ConvNetV1 with the best model in Problem 2.H. Explain the benefit of convolutional neural networks

**Problem 4: Advanced Convolutional Neural Network**

A. Implement ConvNetV2 class in src/models.py
  - ConvNetV2 class has four stages and one output layer.
  - Each stage consists of one downsample block and multiple convolution blocks
  - Downsample blocks and convolution blocks should be grouped by nn.ModuleList
  - Each downsample block consists of Conv2d and BatchNorm2D layers
  - Each convolution block consists of two Conv2d and BatchNorm2D layers and contains residual connection
  - ConvNetV2 class has input arguments “blocks”, “dims” and “strides” are Python List to specify, the number of blocks for each stage, the input and output channels, and the strides of Conv2d layers.
  - Link the output of last stage to the output layer using global average pooling and BatchNorm1D

B. See the model arguments in configs/convnet_v2.yaml. Predict the shape of the output tensor of each stage.

C. Write the performance of ConvNetV2 to Table 1

D. Write the number of trainable parameters of ConvNetV2 to Table 1

E. Compare ConvNetV2 and ConvNetV1. Which model provide better performance? Give a reason why the best model provides higher test accuracy than the other?

**Problem 5: Data Augmentation and Regularization**

A. Explain the below regularization techniques
  - Weight decay
  - Label smoothing
  - Dropout
  - Droppath (a.k.a., stochastic depth)

B. Explain the below data augmentation techniques
  - RandAugment
  - Random Erasing
  - Mixup
  - Cutmix

C. Run ConvNetV3-Reg using configs/convnet_v3_reg.yaml.
   The following settings are already specified in the YAML config:
  - weight_decay: 0.05
  - label_smoothing: 0.1
  - dropout: 0.1
  - droppath: 0.1

   Students should complete the required ConvNetV2 / ConvNetV3-related implementation in src/models.py.

D. Run ConvNetV3-Aug using configs/convnet_v3_aug.yaml.
   The following settings are already specified in the YAML config:
  - weight_decay: 0.05
  - label_smoothing: 0.1
  - dropout: 0.1
  - droppath: 0.1
  - random crop with pad size of 4
  - random horizontal flip with probability of 0.5
  - RandAugment with the number of operation is 2 and magnitude 5
  - Random erasing with probability of 0.5

   Students should implement the data augmentation pipeline in src/data.py.

E. Write the performance of ConvNetV3-Reg and ConvNetV3-Aug to Table 1

F. Write the number of trainable parameters of ConvNetV3-Reg and ConvNetV3-Aug to Table 1

G. Compare ConvNetV2, ConvNetV3-Reg, and ConvNetV3-Aug. Which model provide better test loss? Give a reason why the best model provides lower test loss than the other?

H. Compare ConvNetV2, ConvNetV3-Reg, and ConvNetV3-Aug. Which model provide better performance? Give a reason why the best model provides higher test accuracy than the other?

**Problem 6: Fine-tuning**

A. Explain the architecture of ResNet-50
B. Predict the shape of output tensor of before global pooling
C. Run ResNet50-Scratch and ResNet50-Finetune using configs/resnet50_scratch.yaml and configs/resnet50_finetune.yaml
D. Write the performance of ResNet50-Scratch and ResNet50-Finetune to Table1.
E. Write the number of trainable parameters of ResNet50-Scratch and ResNet50-Finetune to Table 1
F. Compare ResNet50-Scratch and ResNet50-Finetune. Which model provide better performance? Give a reason why the best model provides higher test accuracy than the other?
G. Compare the training settings in configs/resnet50_scratch.yaml and configs/resnet50_finetune.yaml. Give a reason why ResNet50-Finetune uses fewer training epochs and a lower learning rate than ResNet50-Scratch.